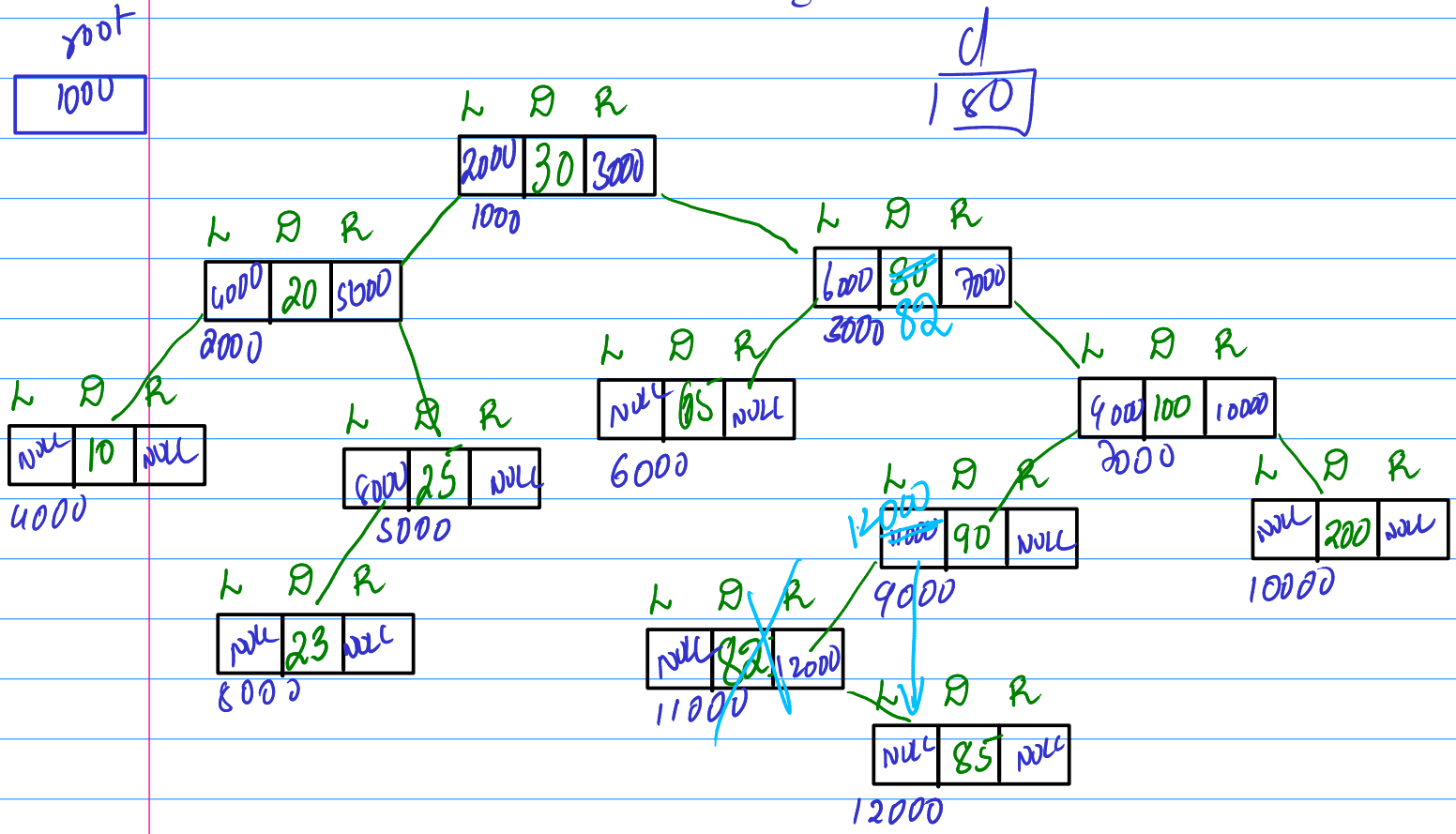
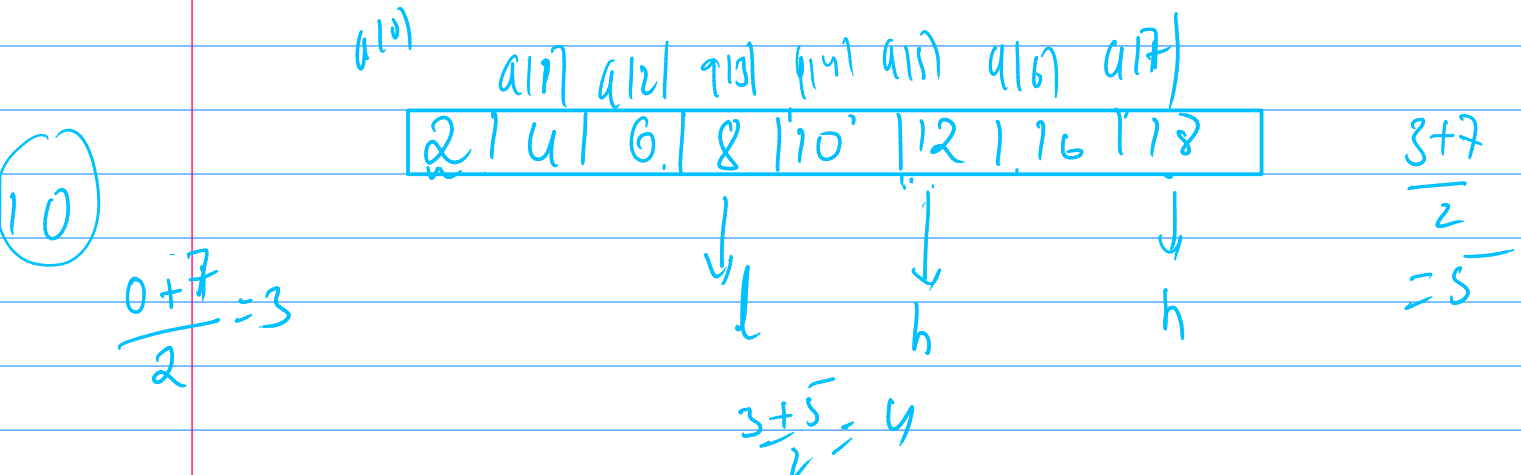
16/04

Data to be deleted is matching with node with 2 children



Binary Search is also known as Non Linear Search



a[0] a[1] a[2] a[3] a[4] a[5] a[6] a[7] a[8]

5	8	1	9	6	7	4	2	3

I n cn

a[0] a[1] a[2] a[3] a[4] a[5] a[6] a[7]

5	1	8	6	7	4	2	3	9

II n-1 c(n-1)

a[0] a[1] a[2] a[3] a[4] a[5] a[6] a[7] a[8]

1	5	6	7	4	2	3	8	9

III n-2 c(n-2)

a[0] a[1] a[2] a[3] a[4] a[5] a[6] a[7] a[8]

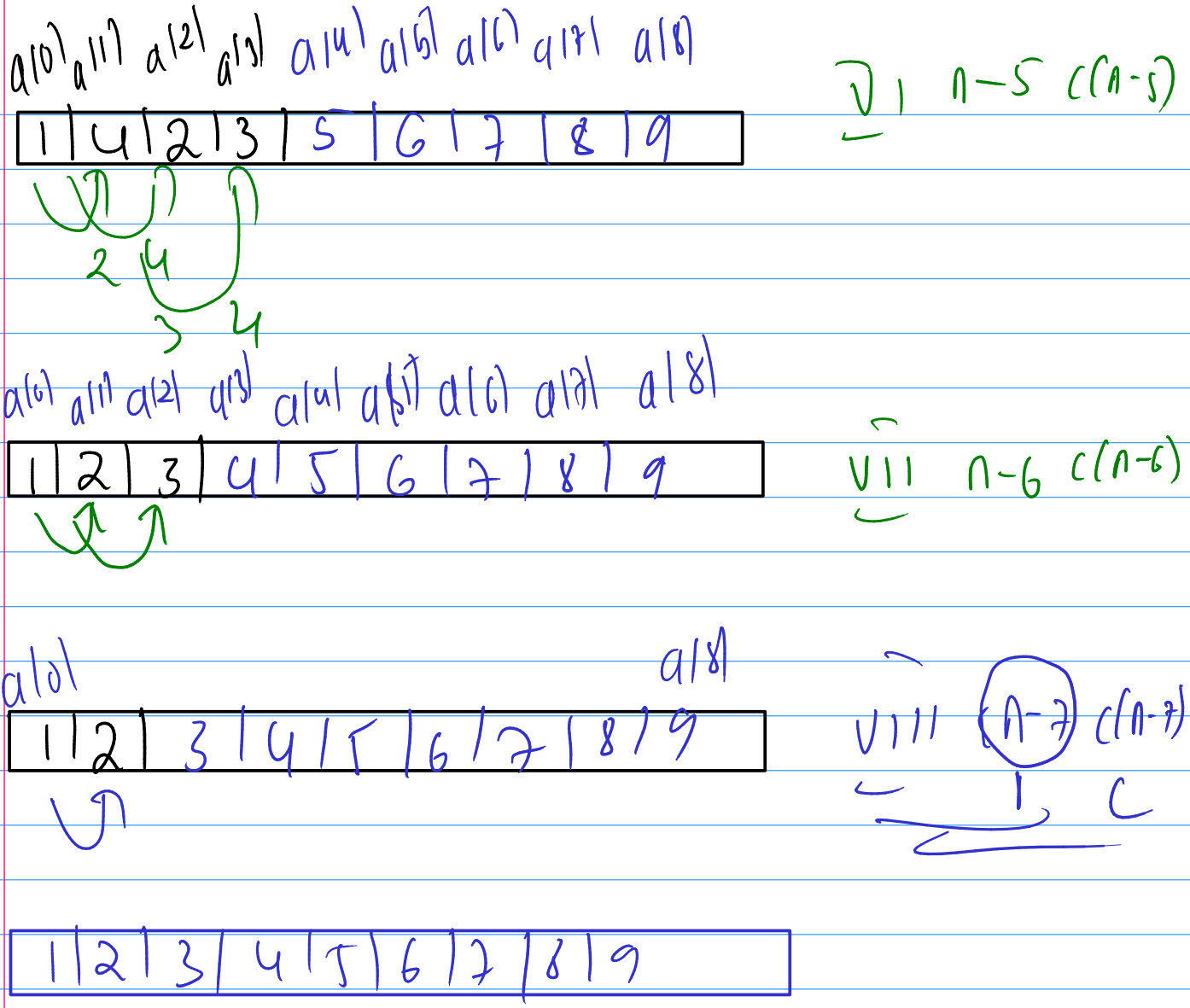
1	5	6	4	2	3	7	8	9

IV n-3 c(n-3)

a[0] a[1] a[2] a[3] a[4] a[5] a[6] a[7] a[8]

1	5	4	2	3	6	7	8	9

V n-4 c(n-4)



total iterations:

cn+c(n-1)+c(n-2)+c(n-3)+c(n-4)+.....+c

c(n+n-1+n-2+n-3+n-4+.....+1)

in the form of arithmetic progressive series

c(n(n-1)/2)

c/2(n(n-1)) → n(n-1)==>n^2-n → low order byte

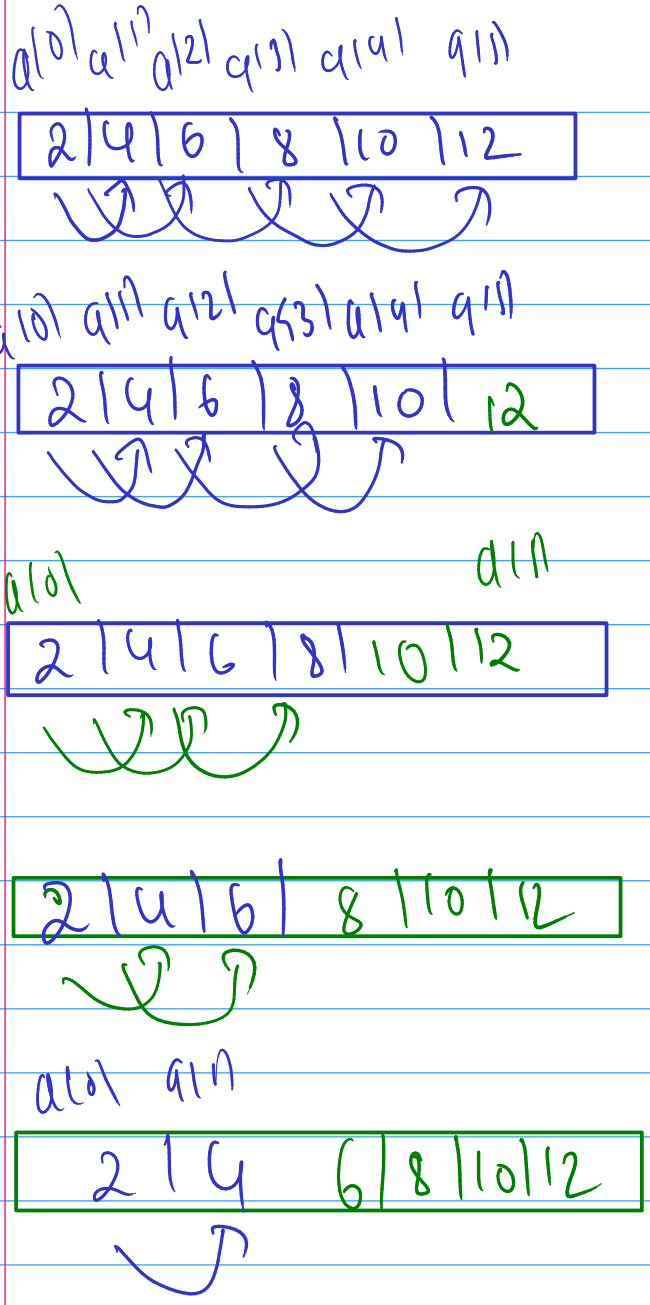
c/2 is constant so we can ignore

n^2 → high order byte

and we can ignore low order byte

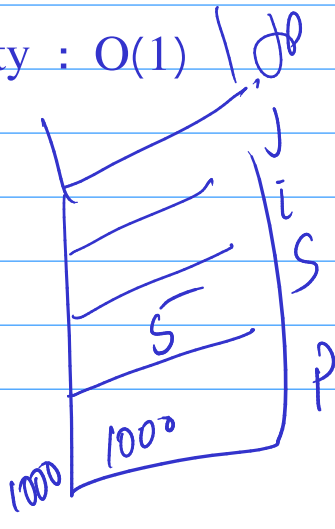
So it is round off to

$O(n^2)$



Time Complexity :
bestcase : $O(n^2)$
average case : $O(n^2)$
worst case : $O(n^2)$

Space Complexity : O(1)




```
void Bubble(int *p, int s)
{ int i, j, temp;
for(i=1; i<size; i++)
{ for(j=0; j<size-1; j++)
{ if(p[j] > p[j+1])
{ temp = p[j];
p[j] = p[j+1];
p[j+1] = temp;
}
}
}
```